

TURARA RYSKULOVA, Kazakhstan

WMO: 383340

Lat: 42.5447N

Lon: 70.3539E

Elev: 2651

StdP: 13.34

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.3	11.6	-2.8	5.2	9.2	1.9	6.7	15.0	23.1	21.5	19.7	23.3	8.2	80	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	95.5	65.9	92.7	64.7	90.2	63.7	68.4	89.1	66.9	87.6	65.5	86.2	6.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	88.4	78.0	59.1	82.5	76.5	57.3	77.4	75.5	34.3	89.6	33.0	87.3	31.9	86.4	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.9	16.5	14.1	DB	2.7	100.1	6.4	2.7	-1.9	102.1	-5.7	103.6	-9.3	105.2	-13.9	107.2
(5)				WB	2.0	72.4	6.3	2.5	-2.6	74.3	-6.3	75.7	-9.8	77.2	-14.4	79.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	31.3	33.3	45.9	55.1	64.7	73.3	77.8	76.5	67.6	55.4	43.6	33.2
	DBStd	18.60	9.73	10.68	10.08	8.57	6.38	5.47	5.18	5.75	7.38	9.34	10.46	10.22
	HDD50	2119	582	474	196	47	1	0	0	1	53	241	524	
	HDD65	4926	1044	888	592	309	85	6	0	2	56	314	645	985
	CDD50	3920	2	6	70	199	458	699	863	821	529	222	48	3
	CDD65	1254	0	0	1	11	77	255	398	358	134	18	2	0
	CDH74	13275	0	0	14	111	661	2586	4634	3910	1204	144	11	0
	CDH80	5665	0	0	0	13	148	1013	2229	1816	416	29	1	0
Wind	WSAvg	5.9	5.9	6.3	6.7	6.7	5.9	5.5	5.4	6.3	5.9	5.2	5.5	5.9
Precipitation	PrecAvg	26.6	3.0	3.2	3.2	3.4	2.7	1.2	0.8	0.4	0.5	2.2	2.8	3.2
	PrecMax	37.8	6.9	8.3	5.6	6.8	4.9	2.7	2.3	1.5	2.9	5.4	6.9	7.0
	PrecMin	18.0	1.4	0.4	1.1	0.6	0.2	0.1	0.0	0.0	0.1	0.2	0.5	
	PrecStd	4.5	1.4	1.5	1.1	1.3	1.2	0.7	0.7	0.4	0.6	1.6	2.0	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.5	64.7	75.6	81.8	87.6	94.8	100.0	99.0	92.2	83.8	74.8	59.4
		MCWB	45.1	48.7	52.0	59.6	64.8	68.2	66.3	66.0	62.6	58.7	54.0	45.0
	2%	DB	51.6	56.5	70.0	77.4	84.1	91.5	96.1	95.6	88.2	77.8	66.8	54.8
		MCWB	41.4	44.8	51.7	59.2	64.0	65.5	66.2	64.9	61.0	55.4	50.6	42.9
	5%	DB	47.5	51.1	64.9	72.9	80.8	88.9	93.1	92.5	85.2	73.6	61.7	49.9
		MCWB	39.5	42.4	50.0	56.6	62.3	64.6	65.1	63.8	59.7	53.8	48.2	40.1
	10%	DB	43.6	47.0	60.5	68.6	77.6	86.1	90.4	89.6	81.2	69.1	57.2	46.1
		MCWB	36.9	40.1	48.7	55.0	60.7	63.7	64.3	62.3	57.7	52.1	46.1	38.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.0	49.4	55.6	63.8	68.9	70.5	70.8	69.2	65.6	59.4	56.5	47.7
		MCDB	56.0	60.5	67.0	76.1	82.8	89.6	91.7	93.3	88.0	80.2	70.1	55.9
	2%	WB	42.8	46.2	53.2	61.0	65.9	68.0	68.4	66.5	62.0	57.1	52.5	44.1
		MCDB	50.1	54.2	65.8	72.9	80.8	87.1	90.8	91.1	85.5	75.0	63.0	51.5
	5%	WB	40.3	43.3	51.2	58.5	63.6	66.5	66.9	64.9	60.1	55.1	49.8	41.6
		MCDB	46.2	50.0	62.5	69.8	78.3	85.4	89.0	89.2	82.7	69.9	60.5	48.4
	10%	WB	37.7	40.5	49.4	56.4	61.6	64.9	65.4	63.4	58.3	53.1	47.1	39.1
		MCDB	42.6	46.6	59.7	66.3	75.5	83.1	87.6	86.8	79.4	66.2	55.1	45.3
Mean Daily Temperature Range	5% DB	MDBR	12.6	13.8	15.9	17.0	19.5	20.8	21.6	21.8	21.3	18.9	15.4	13.2
		MCDBR	16.6	18.3	22.7	23.3	23.1	23.5	24.2	24.3	24.1	23.6	21.3	17.3
		MCWBR	10.2	10.7	11.1	12.9	13.1	12.0	11.3	11.3	11.5	11.7	10.8	10.5
	5% WB	MCDBR	15.3	16.7	19.6	19.6	21.5	21.6	21.8	22.6	22.9	20.8	18.2	15.9
		MCWBR	9.9	10.4	10.7	11.9	13.4	12.3	11.8	11.5	11.7	10.7	10.2	10.3
Clear-Sky Solar Irradiance	taub	0.338	0.381	0.454	0.473	0.451	0.436	0.423	0.391	0.375	0.418	0.396	0.332	
	taud	2.070	1.983	1.890	1.924	2.048	2.122	2.166	2.245	2.241	2.022	2.014	2.089	
	Ebn at Noon	241	250	245	252	263	266	269	274	270	233	215	230	
	Edh at Noon	34	44	56	58	52	49	46	41	38	42	36	31	
All-Sky Solar Radiation	RadAvg	592	858	1138	1542	2007	2318	2338	2138	1677	1019	592	464	
	RadStd	75	113	102	186	156	115	119	87	80	125	89	53	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	+1.19	N/A	N/A	N/A	N/A	N/A	+131

Nomenclature: See separate page